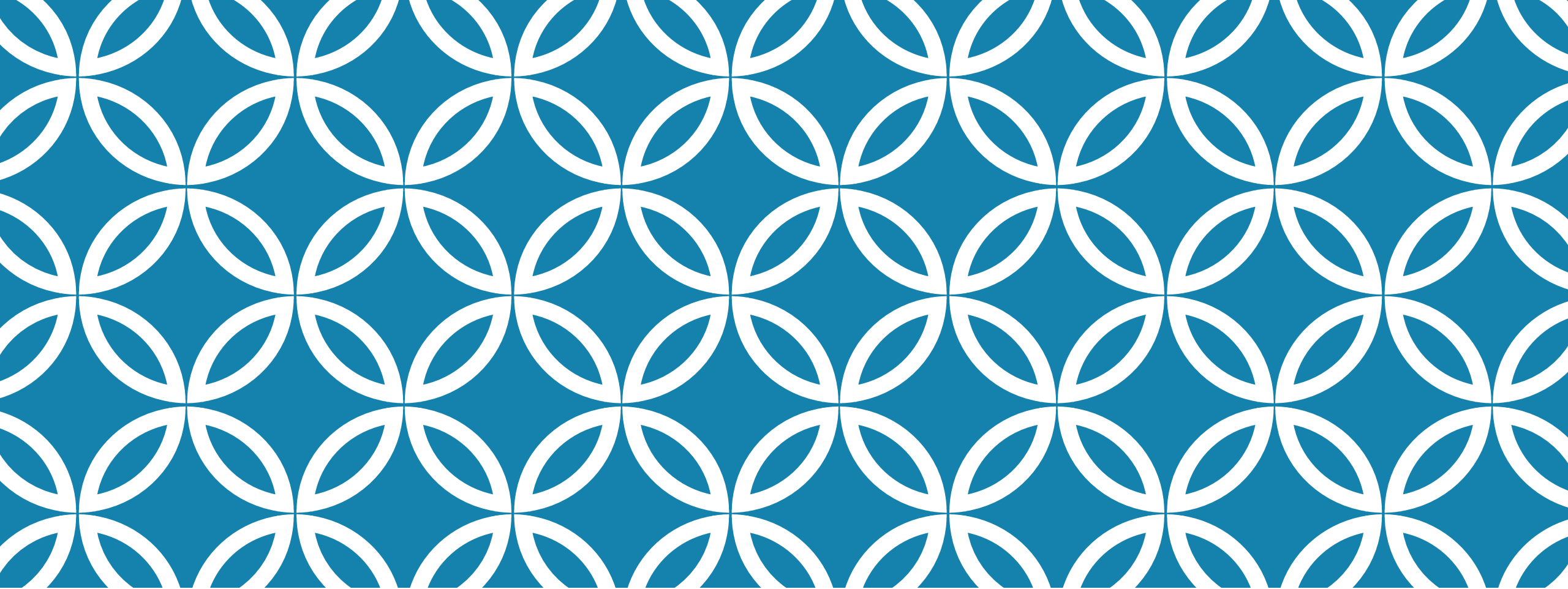
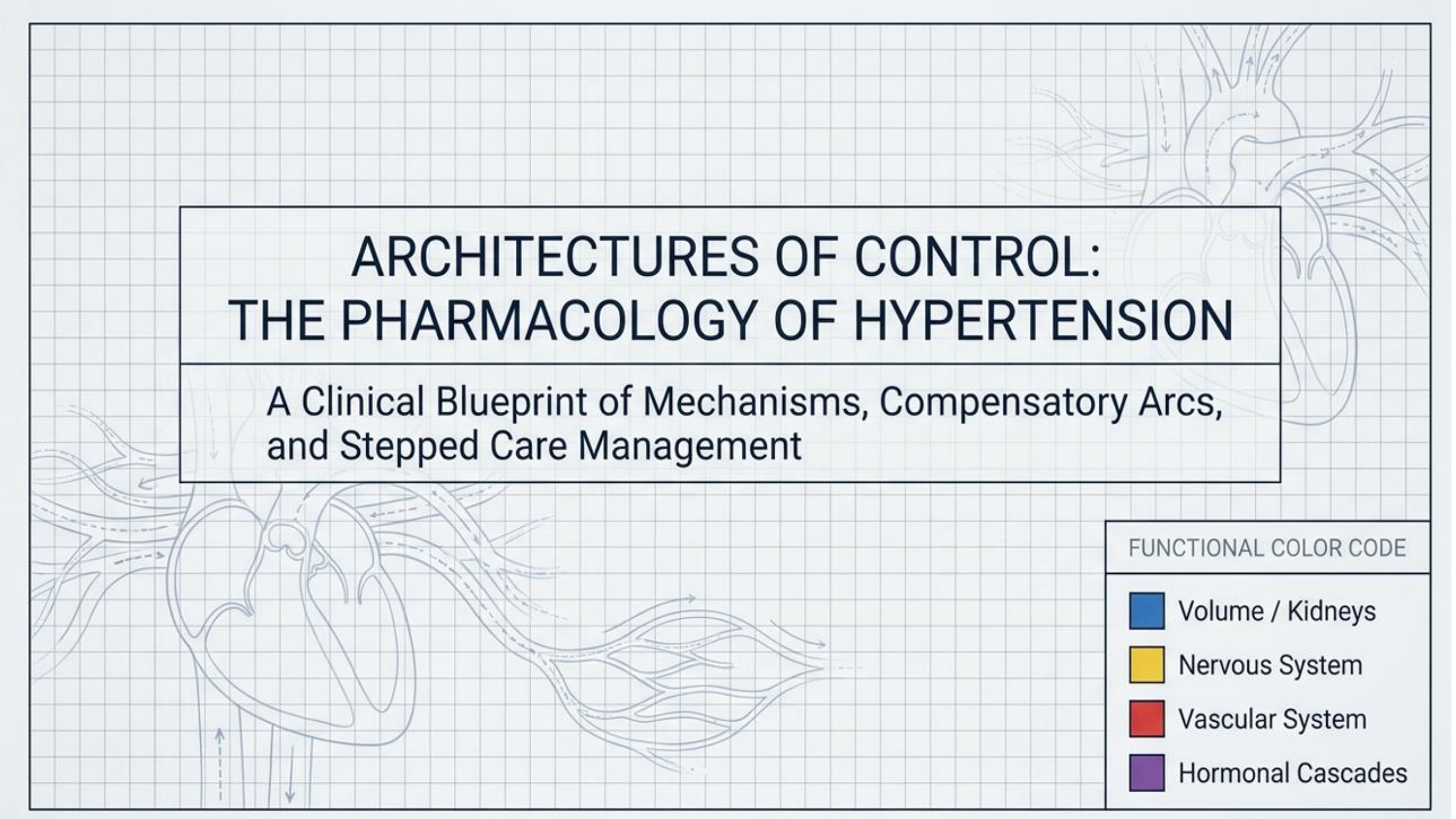




HYPERTENSION

Dr. Doc Tutorial



ARCHITECTURES OF CONTROL: THE PHARMACOLOGY OF HYPERTENSION

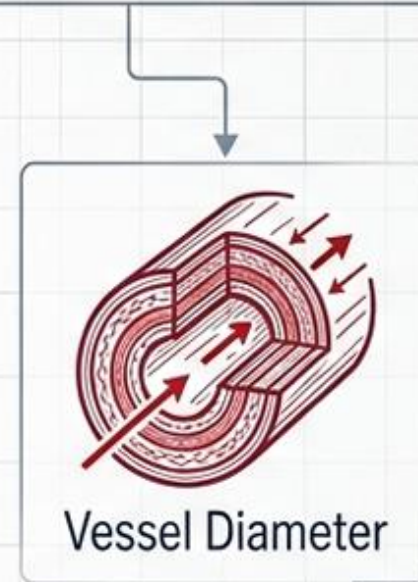
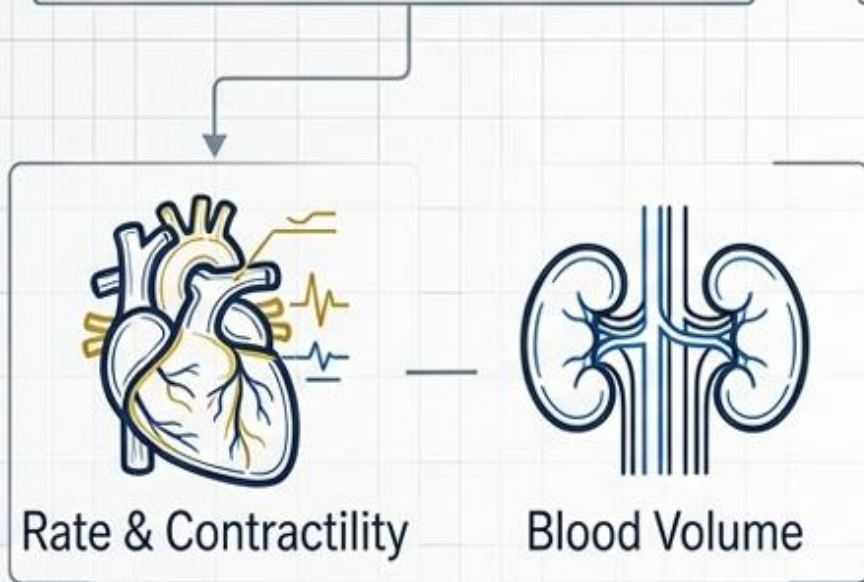
A Clinical Blueprint of Mechanisms, Compensatory Arcs,
and Stepped Care Management

FUNCTIONAL COLOR CODE

-  Volume / Kidneys
-  Nervous System
-  Vascular System
-  Hormonal Cascades

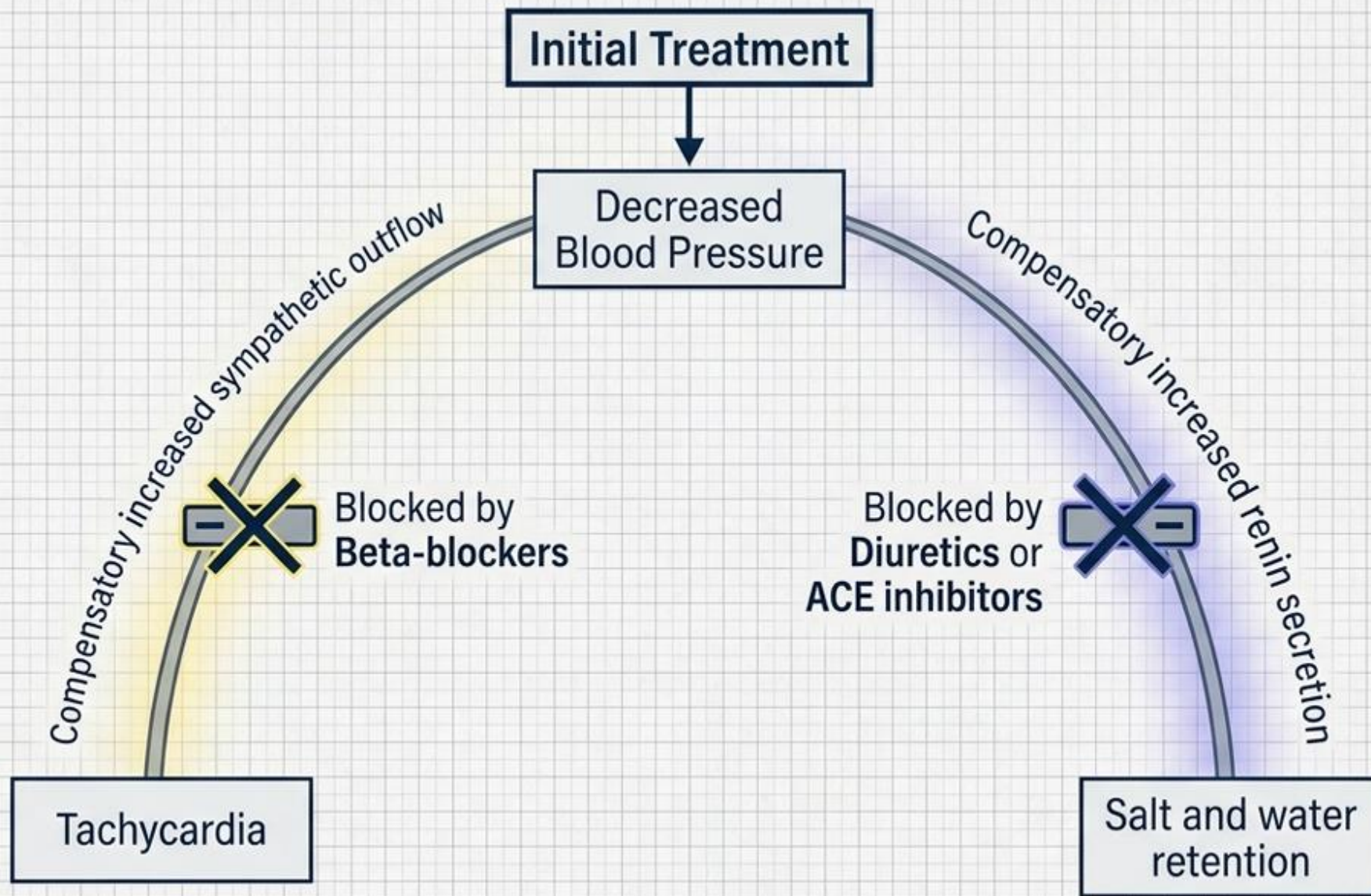
Blood Pressure (BP) =

$$\text{Cardiac Output (CO)} \times \text{Peripheral Vascular Resistance (PVR)}$$



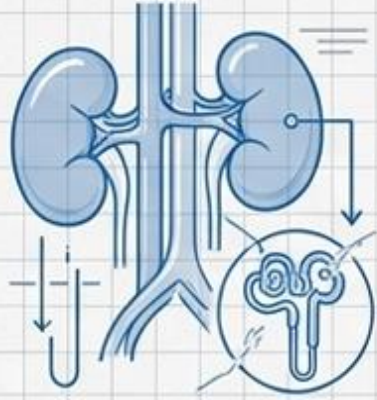
Clinical Insight

Most cases of hypertension are idiopathic (primary/essential). Treatment strategies do not cure the underlying cause; they artificially manipulate this equation by reducing blood volume (blue highlight), sympathetic drive (yellow/gold highlight), vascular tension (red highlight), or angiotensin effects (purple highlight).



The baroreceptor reflex and renin response in primary hypertension are 'reset' to maintain high blood pressure. Therapeutic lowering triggers homeostatic sabotage.

Pillar 1: Diuretics (Volume Reducers)



Mechanism:

Decrease blood volume and alter vascular sodium.

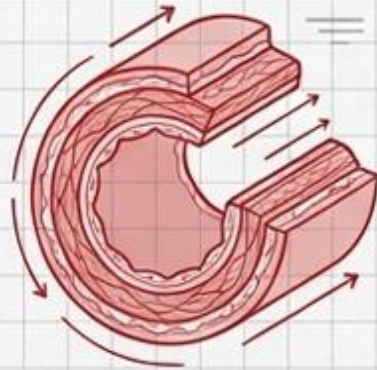
Pillar 2: Sympathoplegics (Nervous System Blockers)



Mechanism:

Reduce sympathetic outflow and cardiac/vascular stimulation.

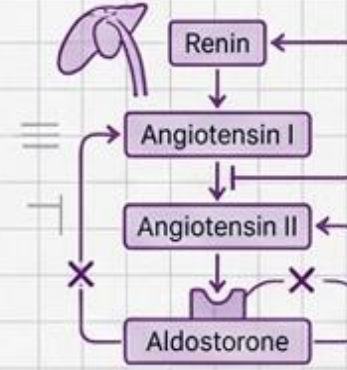
Pillar 3: Vasodilators (Vessel Relaxers)



Mechanism:

Directly relax smooth muscle tension.

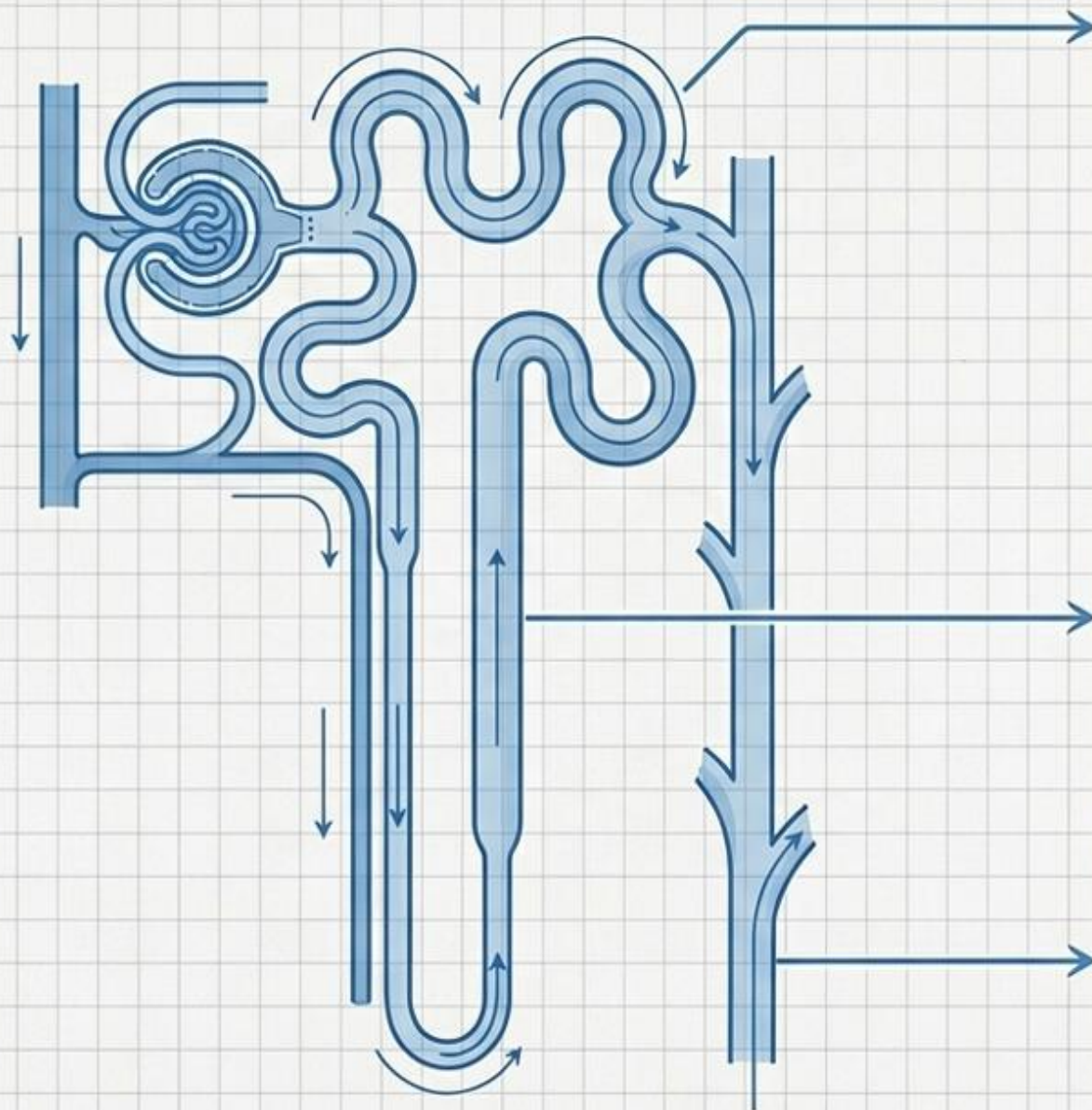
Pillar 4: RAAS Inhibitors (Hormonal Interrupters)



Mechanism:

Block angiotensin II vasoconstriction and aldosterone-mediated retention.

Pillar 1: Diuretics - The Nephron Map



Distal Convoluted Tubule

Drugs: **Thiazides** (Hydrochlorothiazide, Chlorthalidone)

Mechanism: Block Na/Cl transporter.

Clinical: 1st line for mild/moderate hypertension. Maximal effect at low doses.

Toxicity: Hypokalemia, hyperuricemia (avoid in gout), hyperglycemia.

Thick Ascending Limb

Drugs: **Loop Diuretics** (Furosemide)

Mechanism: Block Na/K/2Cl transporter. High-ceiling diuresis.

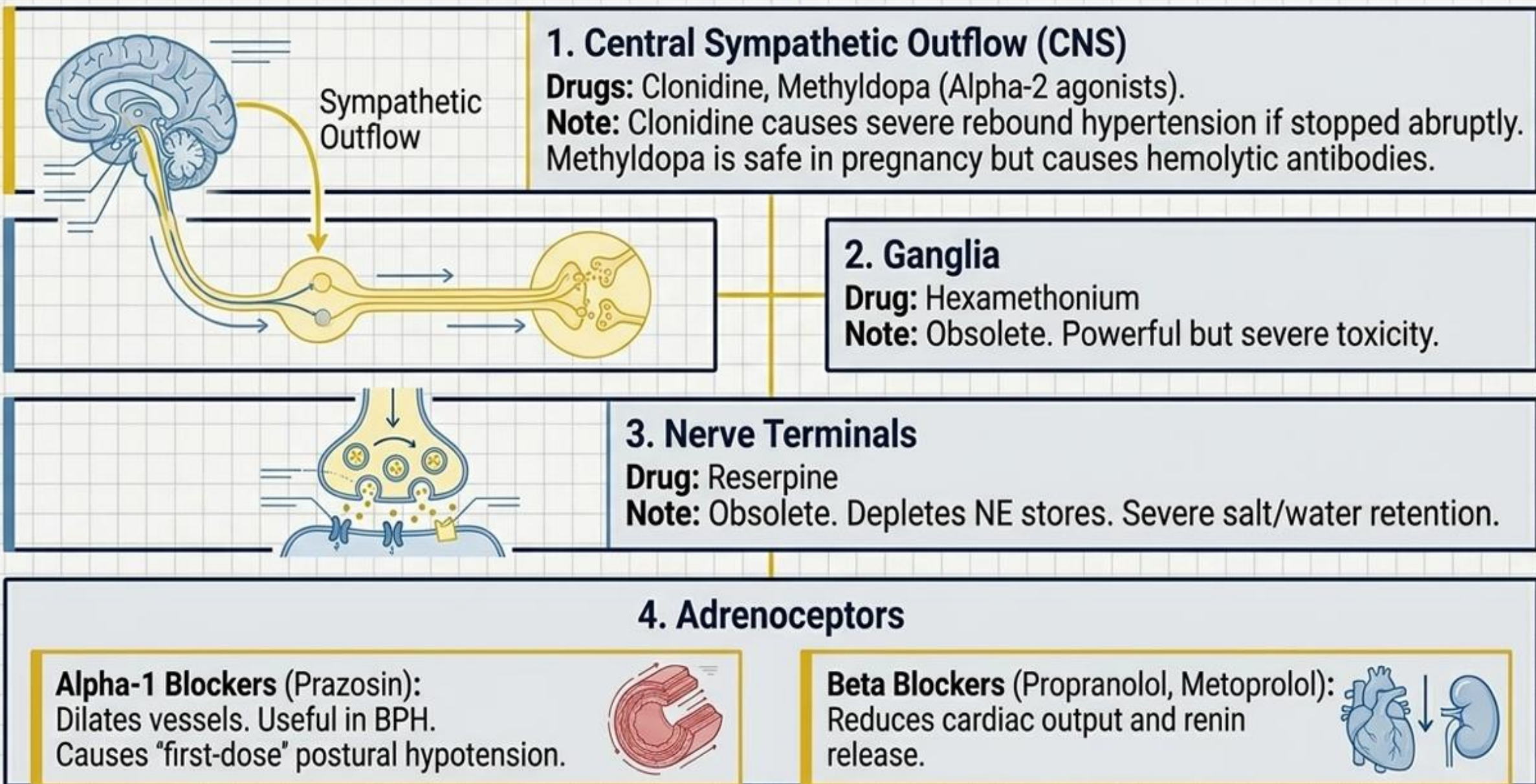
Clinical: Reserved for severe hypertension, CHF, and renal dysfunction. Not for mild HTN.

Collecting Duct

Drugs: **Potassium-sparing agents** (Spironolactone)

Clinical: Used primarily in combination to offset hypokalemia.

Pillar 2: Sympathoplegics - Sites of Action



The Beta-Blocker Spectrum

Beta-1 Selective (Cardioselective)

Drugs: Metoprolol, Atenolol, Esmolol.

Profile: Target heart specifically. Safer in patients with asthma, diabetes, or peripheral vascular disease (avoids blocking beneficial Beta-2 vasodilation/bronchodilation).



Non-Selective (Beta-1 + Beta-2)

Drugs: Propranolol.

Toxicity Risk: Can cause bronchospasm, mask hypoglycemia, and increase hyperlipidemia.



Combined Alpha and Beta Blockers

Drugs: Labetalol, Carvedilol.

Clinical: Labetalol used in hypertensive emergencies and pregnancy. Carvedilol reduces mortality in CHF.



KEY SHIFT ALERT:

JNC-VIII guidelines no longer consider Beta-blockers as first-line for uncomplicated hypertension due to cardiovascular outcome data.

Pillar 3: Vasodilators - Mechanisms of Relaxation

Mechanism 1: Nitric Oxide Release

Drugs: Nitroprusside, Hydralazine.

Notes: Hydralazine requires intact endothelium; subject to fast/slow acetylator genetics.

Mechanism 2: Potassium Channel Opening

Drugs: Minoxidil, Diazoxide.

Notes: Hyperpolarizes smooth muscle. Minoxidil causes severe reflex tachycardia requiring Beta-blocker co-administration.

Mechanism 3: Dopamine D1 Activation

Drug: Fenoldopam. **Notes:** Rapid IV infusion for emergencies; maintains renal perfusion.

Mechanism 4: Calcium Channel Blockade

(See next slide for full breakdown).

Class-Wide Toxicity Warning: All direct vasodilators trigger compensatory reflex tachycardia and fluid retention. Best utilized with diuretics and Beta-blockers.

Calcium Channel Blockers: The Core Distinction

Dihydropyridines (DHPs)



Target: Vessels > Heart (Peripherally acting).

Drugs: Amlodipine, Nifedipine, Nicardipine.

→ **Effects:** Potent vasodilation. May cause reflex tachycardia.

→ **Clinical:** Safe in pregnancy. First-line for general hypertension. (Note: Sublingual nifedipine banned due to MI risk).

Non-Dihydropyridines



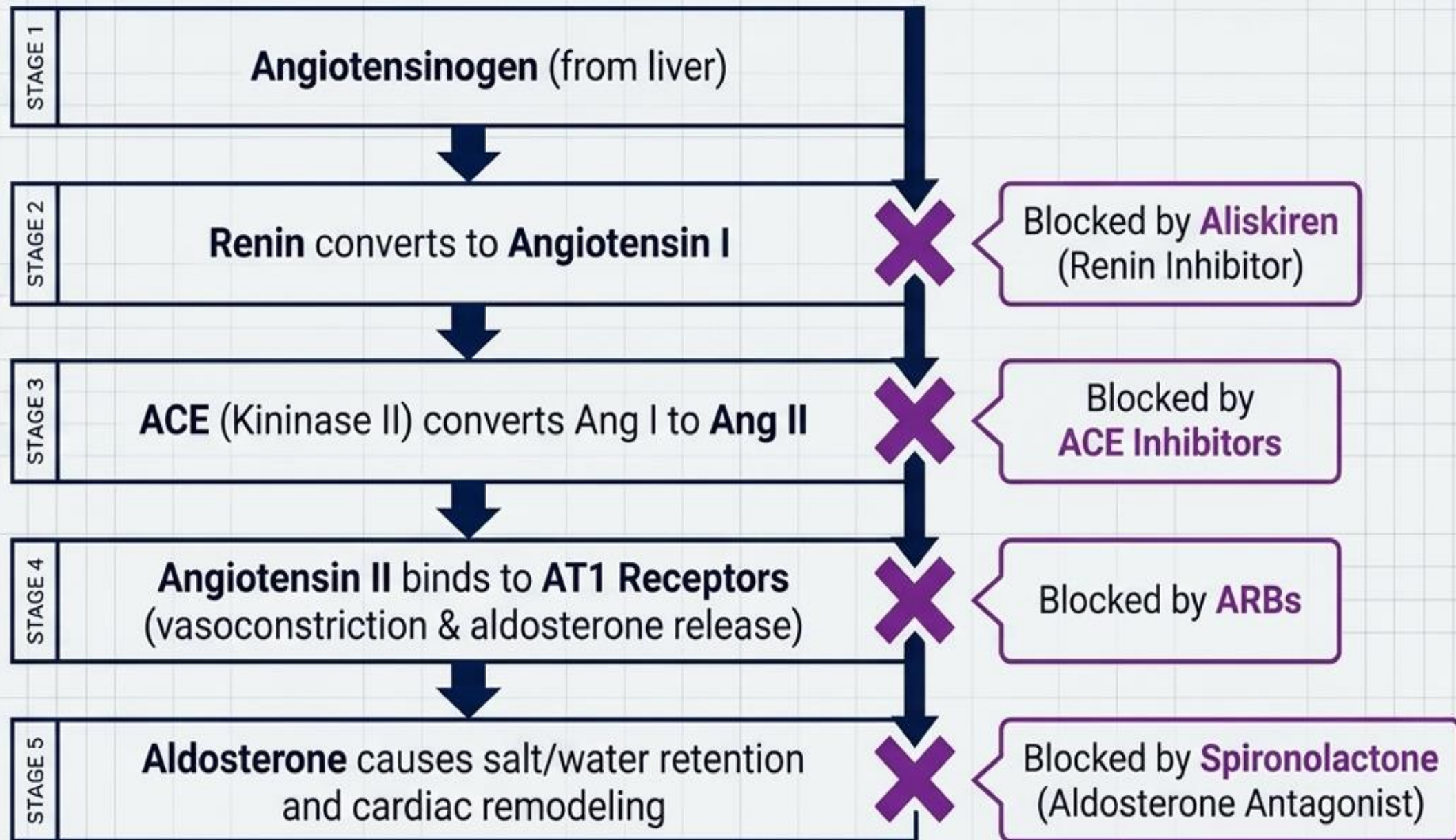
Target: Heart ≥ Vessels.

Drugs: Verapamil, Diltiazem.

→ **Effects:** Direct cardiodepressant activity (decreases HR, contractility, AV conduction).

→ **Contraindications:** Avoid in CHF, Sick Sinus Syndrome, or alongside Beta-blockers (severe cardiac depression).

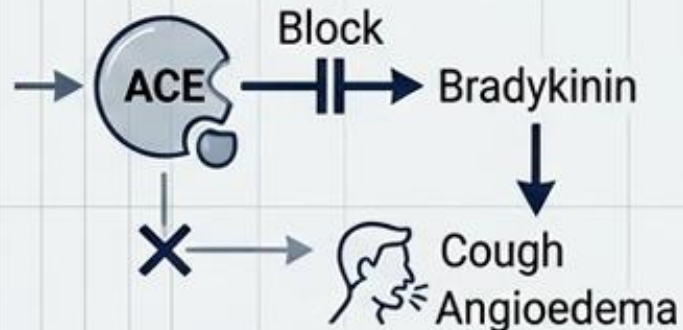
Pillar 4: RAAS Inhibitors - The Hormonal Cascade



Diagnostic Comparison: ACEi vs. ARBs

ACE Inhibitors (Captopril, Enalapril)

- **Unique Mechanism:** Inhibits ACE (Kininase II), which also normally breaks down Bradykinin.
- **Unique Toxicity:** Bradykinin accumulation causes dry cough (up to 30% of patients) and life-threatening angioedema.



Shared Traits (The Overlap)

Highly effective in diabetes (renal protection) and heart failure (prevents remodeling).

Shared Toxicities: Hyperkalemia, postural hypotension in volume-depleted states.

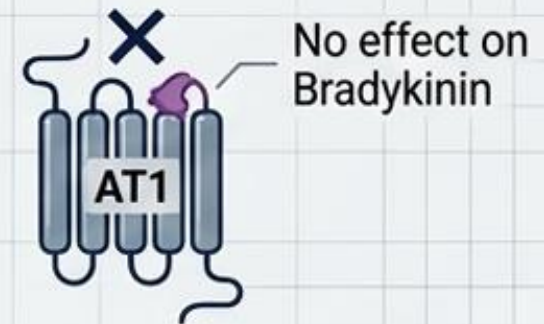
ABSOLUTE CONTRAINDICATION: Teratogenic (fetal renal damage)—strictly prohibited in pregnancy. Do NOT prescribe ACEi and ARB simultaneously.



**NO
COMBINATION**

ARBs (Losartan, Valsartan)

- **Unique Mechanism:** Directly blocks AT1 receptors. No effect on bradykinin.
- **Advantage:** Lower incidence of cough and angioedema.



Guideline Evolution: JNC-VII to JNC-VIII

JNC-VII Paradigm

Goal: <140/90 for all.

First-line: **Thiazides** preferred. 

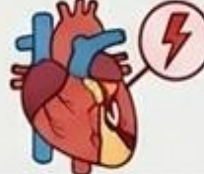
Beta-blockers included in first-line arsenal. 

Evolution of
Standard of Care

JNC-VIII Paradigm Shift

- **Age-Based Goals:** <140/90 for ages <60 (and all patients with CKD/DM). Relaxed to <150/90 for patients >60 without CKD/DM. 
- **First-Line Arsenal:** **Thiazides**, **ACEi inhibitors**, **ARBs**, **CCBs**. 
RAAS
↓
cascade
- **The Demotion:** **Beta-blockers** are NO LONGER first-line therapies.  
- **Ethnicity Factor:** In patients of African descent without CKD, **CCBs** or **Thiazides** are preferred over **ACEi/ARBs**. In CKD patients of any race, **ACEi/ARBs** are first choice. RAAS
↓
kidney

The Clinical Matching Matrix (Stepped Care Architecture)

Comorbidity	Preferred Choice(s)	Contraindicated / Avoid
 Diabetes / CKD 	ACEi, ARB  RAAS ↓ cascade (protects diabetic kidney)	Beta-blockers , high-dose  Thiazides   (metabolic impairment)
Benign Prostatic Hyperplasia (BPH) 	Alpha-1 blockers  (Prazosin)	
Heart Failure (CHF) 	ACEi, ARB, Beta-blockers  RAAS ↓ cascade (Carvedilol),   Spironolactone	Non-DHP CCBs   (Verapamil)
Pregnancy 	Methyldopa, Labetalol, Nifedipine  	RAAS ↓ cascade ACEi, ARBs (Teratogenic) 
Ischemic Heart Disease (Angina) 	 Beta-blockers, CCBs 	Direct vasodilators   (cause reflex tachycardia/increased cardiac work) 



Emergency Protocol: Hypertensive Crisis

Definition & Goal

Accelerated severe hypertension associated with **rapidly progressing end-organ damage**.

Goal: Promptly lower BP to 140-160/90-110 within hours, then proceed slowly via IV administration.

The IV Arsenal



Sodium Nitroprusside



Powerful **combined arteriolar** and venodilator.
(Used with Esmolol in aortic dissection).



Nicardipine



Longest acting parenteral CCB;
Drug of Choice for many emergencies.



Fenoldopam



D1 agonist, preserves renal perfusion.



Labetalol



Combined Alpha/Beta blocker.
Drug of Choice for emergencies in pregnancy.

Mechanism Note

These powerful vasodilators are often combined with **loop diuretics (Furosemide)** and Beta-blockers to suppress the massive reflex tachycardia and fluid retention they trigger.

